

ST6101: Advanced Statistical Theory (I)

Lecture 10: Consistency of MLE and M -Estimators

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*Adapted from Dr. Dongming Huang's slides.

Recap: Method of Moments

Suppose X_i 's are i.i.d. from P_θ , $\theta \in \Theta \subset \mathbb{R}^k$ and $E_\theta |X_1|^k < \infty$

- Let $\mu_j = E_\theta X_1^j$ be the j th moment of P_θ and suppose $\mu_j = h_j(\theta)$ for some functions h_j on $\mathcal{R}^k$ ($j = 1, \dots, k$)
- $\hat{\mu}_{n,j} = \frac{1}{n} \sum_{i=1}^n X_i^j$ is the j th sample moment
- Any $\hat{\theta}_n$ that solves

$$\hat{\mu}_{n,j} = h_j(\hat{\theta}_n), \quad j = 1, \dots, k$$

is a moment estimator of θ

Properties of MOM Estimators

Let $\mu = (\mu_1, \dots, \mu_k)$, $\hat{\mu}_n = (\hat{\mu}_{n,1}, \dots, \hat{\mu}_{n,k})$, and $h = (h_1, \dots, h_k)$. Then

$$\mu = h(\theta) \text{ and } \hat{\mu}_n = h(\hat{\theta}_n)$$

- If h^{-1} exists, the unique moment estimator of θ is $\hat{\theta}_n = h^{-1}(\hat{\mu}_n)$
- Furthermore, if h^{-1} continuous, then by SLLN and continuous mapping, $\hat{\theta}_n$ is strongly consistent
- If $g = h^{-1}$ is differentiable and $E|X_1|^{2k} < \infty$, by CLT and δ -method, we have

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{\mathcal{D}} N(0, [\nabla g]^\top V_\mu \nabla g)$$

where V_μ is a $k \times k$ matrix whose (i, j) th element is $\mu_{i+j} - \mu_i \mu_j$

- In this case, the MOM estimator is $\sqrt{n}$ -consistent
- If $k = 1$, $\text{amse}_{\hat{\theta}_n}(\theta) = g'(\mu_1)^2 \sigma^2 / n$ where $\sigma^2 = \mu_2 - \mu_1^2$

Example 3.24

Let $X_1, \dots, X_n$ be i.i.d. from P_θ indexed by $\theta = (\mu, \sigma^2)$, where $\mu = EX_1 \in \mathcal{R}$ and $\sigma^2 = \text{Var}(X_1) \in (0, \infty)$.

This includes cases like normal distributions and **double exponential** distributions with Lebesgue p.d.f.

$$\frac{1}{\sqrt{2}\sigma} e^{-\sqrt{2}|x-\mu|/\sigma}, \quad \mu \in \mathbb{R}, \sigma > 0$$

- Since $EX_1 = \mu$ and $EX_1^2 = \text{Var}(X_1) + (EX_1)^2 = \sigma^2 + \mu^2$, setting $\hat{\mu}_1 = \mu$ and $\hat{\mu}_2 = \sigma^2 + \mu^2$ we obtain the moment estimator

$$\hat{\theta} = \left(\bar{X}, \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \right) = \left(\bar{X}, \frac{n-1}{n} S^2 \right).$$

Note that $\bar{X}$ is unbiased, but $\frac{n-1}{n} S^2$ is not.

- If P_θ is normal, then $\hat{\theta}$ is sufficient and complete, and is nearly the same as the UMVUE
- If P_θ is double exponential, then $\hat{\theta}$ is not sufficient and can often be improved

Consider now the estimation of σ^2 when we know that $\mu = 0$.

- Obviously we cannot use the equation $\hat{\mu}_1 = \mu$ to solve the problem.
- Using $\mu_2 = \sigma^2$, we obtain the moment estimator

$$\hat{\sigma}^2 = \hat{\mu}_2 = n^{-1} \sum_{i=1}^n X_i^2. \quad (1)$$

- If P_θ is normal, this is a good estimator
- If P_θ is double exponential, Eq (1) is not sufficient for σ .
 - We should first make a transformation $Y_i = |X_i|$ (note that $EY_i = \sigma/\sqrt{2}$),
 - then obtain the moment estimator based on the transformed data:

$$2\bar{Y}^2 = 2 \left(\frac{1}{n} \sum_{i=1}^n |X_i| \right)^2,$$

which is sufficient for σ^2 .

Asymptotic Properties of UMVUE

- UMVUE's are typically consistent
- UMVUE's are exactly unbiased so that there is no need to discuss their asymptotic biases
- The amse can be used to assess the performance of a UMVUE if the exact form of mse are difficult to obtain
- In many cases, although the variance does not attain the Cramér-Rao lower bound, the ratio of the mse over the Cramér-Rao lower bound converges to 1

Example: Normal Families

Model

$X_1, \dots, X_n$ be i.i.d. from the $N(\mu, \sigma^2)$ distribution with an unknown $\mu \in \mathcal{R}$ and a known σ^2 . $I(\mu) = n/\sigma^2$. We are interested in estimating μ^2

- We can show that $h(\bar{X}) = \bar{X}^2 - \sigma^2/n$ is the UMVUE of μ^2 and the variance is

$$\text{Var}(h(\bar{X})) = \frac{4\mu^2\sigma^2}{n} + \frac{2\sigma^4}{n^2}$$

while the Cramér-Rao lower bound is $4\mu^2\sigma^2/n$.

- If $\mu = 0$, the C-R lower bound $= 0$ and is not informative
- If $\mu \neq 0$, the ratio of the variance of the UMVUE and the C-R lower bound is

$$1 + \frac{2\sigma^4}{n^2} \times \frac{n}{4\mu^2\sigma^2} \rightarrow 1$$

Example: Poisson Families

Model

Let $X_1, \dots, X_n$ be i.i.d. from the $\text{Poisson}(\lambda)$ distribution with an unknown $\lambda > 0$. We are interested in estimating $\tau = P(X_1 = 0) = e^{-\lambda}$.

- Have showed $T_n = \sum_{i=1}^n X_i$ is sufficient and complete. Since $S_0 = I_{\{X_1=0\}}$ is unbiased, by Lehmann-Scheffé Theorem

$$S_n = E(S_0 | T_n) = \left(1 - \frac{1}{n}\right)^{T_n}$$

is a UMVUE

- $\log(S_n) = T_n/n [n \log(1 - 1/n)] = T_n/n [-1 + O(1/n)] = -T_n/n + O_p(1/n)$ since $T_n/n = O_p(1)$ by SLLN
- By CLT, $\sqrt{n}(T_n/n - \lambda) \xrightarrow{\mathcal{D}} N(0, \lambda)$
- By Slutsky's theorem and $O_p(1/\sqrt{n}) = o_p(1)$, we have $\sqrt{n}(\log S_n - (-\lambda)) \xrightarrow{\mathcal{D}} N(0, \lambda)$
- By δ -method, $\sqrt{n}(S_n - e^{-\lambda}) \xrightarrow{\mathcal{D}} N(0, \lambda e^{-2\lambda})$.
So $\text{amse}_{S_n}(\lambda) = \lambda e^{-2\lambda}/n$; equals to C-R lower bound.

Asymptotic Property of Sample Quantiles

For any $\gamma \in (0, 1)$, the $[\gamma n]$ th order statistic is also called the γ -sample quantile. ($[a]$ denotes the largest integer that is no greater than a)

Theorem

- 1 Let $X_1, X_2, \dots$ be i.i.d. r.v.s with CDF F ,
- 2 let $\gamma \in (0, 1)$, and
- 3 let $\tilde{\theta}_n$ be the $[\gamma n]$ th order statistic for $X_1, \dots, X_n$

Suppose $F(\theta) = \gamma$ and $F'(\theta)$ exists and is positive, then

$$\sqrt{n}(\tilde{\theta}_n - \theta) \xrightarrow{\mathcal{D}} N\left(0, \frac{\gamma(1-\gamma)}{[F'(\theta)]^2}\right)$$

- This shows that if the CDF is smooth and strictly increasing around θ , then the $F(\theta)$ -sample quantile is asymptotically unbiased, and its amse is $\gamma(1-\gamma)/(nF'(\theta)^2)$

- For any fixed $a \in \mathbb{R}$, define

$$S_n(a) = \# \{i \leq n : X_i \leq a\}$$

- $\tilde{\theta}_n \leq a \Leftrightarrow S_n(a) \geq \lfloor \gamma n \rfloor$
- Therefore for any fixed $t \in \mathbb{R}$

$$P\left(\sqrt{n}(\tilde{\theta}_n - \theta) \leq t\right) = P\left(S_n(\theta + t/\sqrt{n}) \geq \lfloor \gamma n \rfloor\right).$$

- $S_n(a) \sim \text{Binom}(n, F(a))$, so it is tempting to use CLT.
- However, CLT is not precise enough ($F_n(x) \rightarrow F(x)$ for a given x).
- We need a uniform control on the convergence of CDF:

Berry-Esseen Theorem .

Theorem (Berry–Esseen Theorem)

There exists a positive universal constant C such that the following holds. Suppose $Y_1, Y_2, \dots, Y_n$ are i.i.d. random variables with $E(Y_1) = 0$, $E(Y_1^2) = \sigma^2 > 0$, and $E(|Y_1|^3) = \rho < \infty$. Let F_n be the CDF of

$$\frac{\sum_{i=1}^n Y_i}{\sigma\sqrt{n}}$$

and Φ be the CDF of the standard normal distribution, then

$$\sup_{y \in \mathbb{R}} |F_n(y) - \Phi(y)| \leq \frac{C\rho}{\sigma^3\sqrt{n}} \quad .$$

Proof (Cont.)

- Let $u = \theta + t/\sqrt{n}$. Then $S_n(\theta + t/\sqrt{n}) = \sum_i I_{X_i \leq u}$.
- Consider $Y_{n,i} = I_{X_i \leq u} - F(u)$. Then $EY_{n,i} = 0$, and $E|Y_{n,i}|^3 \leq 1$.
- For all n large enough, so that u is in a fixed neighbor of θ , we have $EY_{n,i}^2 = F(u)(1 - F(u)) \geq \sigma_0^2 > 0$.
- By Berry–Esseen Theorem,

$$\begin{aligned} & \left| P\left(S_n(\theta + t/\sqrt{n}) \geq \lfloor \gamma n \rfloor\right) - \Phi\left(-\frac{\lfloor \gamma n \rfloor - nF(u)}{\sqrt{nF(u)(1 - F(u))}}\right) \right| \\ & \leq \frac{C}{\sigma_0^3 \sqrt{n}} \end{aligned}$$

- Note that

$$\begin{aligned}
 & \lim_n \frac{[\gamma n] - nF(\theta + t/\sqrt{n})}{\sqrt{nF(\theta + t/\sqrt{n})(1 - F(\theta + t/\sqrt{n}))}} \\
 &= \frac{1}{\sqrt{F(\theta)(1 - F(\theta))}} \lim_n \sqrt{n} ([\gamma n]/n - \gamma + \gamma - F(\theta + t/\sqrt{n})) \\
 &= \frac{1}{\sqrt{\gamma(1 - \gamma)}} (-F'(\theta)t)
 \end{aligned}$$

- Since Φ is continuous, we conclude that for any $t \in \mathbb{R}$

$$\lim_n \left| P(\sqrt{n}(\tilde{\theta}_n - \theta) \leq t) - \Phi\left(\frac{1}{\sqrt{\gamma(1 - \gamma)}} F'(\theta)t\right) \right| = 0$$

- Since this holds for any $t \in \mathbb{R}$, we conclude that

$$\sqrt{n}(\tilde{\theta}_n - \theta) \xrightarrow{\mathcal{D}} N\left(0, \frac{\gamma(1 - \gamma)}{[F'(\theta)]^2}\right)$$

Wald's Consistency of MLE: Basic Idea

Let $X_1, \dots, X_n$ be i.i.d. with density $f(x | \theta)$ w.r.t. a σ -finite measure ν , where $\theta \in \Theta$. θ_0 is the true value.

- The MLE is the value of θ that maximizes

$$\begin{aligned}\ell_n(\theta) - \ell_n(\theta_0) &= \log L_n(\theta) - \log L_n(\theta_0) \\ &= \sum_{j=1}^n (\log f(X_j | \theta) - \log f(X_j | \theta_0))\end{aligned}$$

- From the Strong Law of Large Numbers,

$$\frac{1}{n} \log \frac{L_n(\theta)}{L_n(\theta_0)} = \frac{1}{n} \sum_{j=1}^n \log \frac{f(X_j | \theta)}{f(X_j | \theta_0)} \xrightarrow{\text{a.s.}} E_{\theta_0} \log \frac{f(X | \theta)}{f(X | \theta_0)}$$

- Suppose the model is identifiable
- If $\theta = \theta_0$, RHS = 0. If $\theta \neq \theta_0$, then RHS < 0.
- Moreover, if the model is **well-separated**, i.e., for any $\|\theta - \theta_0\| \geq \rho$, the RHS < $-\delta_\rho < 0$, then eventually, $\frac{L_n(\theta)}{L_n(\theta_0)} < \exp(-n\delta_\rho)$, and the MLE must be lied in $B(\theta_0, \rho)$

- A defect in the last argument: The SLLN only applies to a fixed θ , from which we cannot conclude that $\frac{L_n(\theta)}{L_n(\theta_0)}$ is uniformly small on $\|\theta - \theta_0\| \geq \rho$
- To make it precise, we need a Uniform SLLN

Theorem (USLLN, C)

Let $X_1, \dots, X_n, \dots$, be i.i.d. sample from P on the space $\mathcal{X}$. Let $U(x, \theta)$ be a measurable function on $\mathcal{X} \times \Theta$. Suppose

- 1 $U(x, \theta)$ is a continuous in θ for any fixed x ,
- 2 for each θ , $\mu(\theta) = EU(X, \theta)$ is finite,
- 3 Θ is compact,
- 4 there exists a function $M(x)$ such that $EM(X) < \infty$ and $|U(x, \theta)| \leq M(x)$ for all x and θ .

Then

$$P \left\{ \lim_{n \rightarrow \infty} \sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{j=1}^n U(X_j, \theta) - \mu(\theta) \right| = 0 \right\} = 1$$

- By the continuity of $U(x, \cdot)$, its boundedness by $M(x)$, and DCT, $\mu(\theta)$ is continuous.

W.L.O.G., assume $\mu(\theta) \equiv 0$; otherwise consider

$$\tilde{U}(x, \theta) = U(x, \theta) - \mu(\theta)$$

- Let

$$\varphi(x, \theta, \rho) = \sup_{\|\theta' - \theta\| < \rho} U(x, \theta')$$

- φ is measurable, bounded by M and $\varphi(x, \theta, \rho) \rightarrow U(x, \theta)$ as $\rho \rightarrow 0$
- Fixed $\varepsilon > 0$. For each θ , find $\rho_\theta > 0$ so that $E\varphi(x, \theta, \rho_\theta) < \varepsilon$.
 - By DCT, $E\varphi(X, \theta, \rho) \rightarrow EU(X, \theta) = \mu(\theta) = 0$, as $\rho \rightarrow 0$
- Note that the collection of $B(\theta, \rho_\theta) = \{\theta' : |\theta - \theta'| < \rho_\theta\}$ for all θ covers Θ .

By the compactness of Θ , there exists a finite sub-cover, say,

$$\Theta \subset \bigcup_{j=1}^m B(\theta_j, \rho_{\theta_j})$$
- For each $\theta \in \Theta$, there exists an index j such that $\theta \in B(\theta_j, \rho_{\theta_j})$ and $U(x, \theta) \leq \varphi(x, \theta_j, \rho_{\theta_j})$ for all x .

- So

$$\sup_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n U(X_i, \theta) \leq \max_{1 \leq j \leq m} \frac{1}{n} \sum_{i=1}^n \varphi(X_i, \theta_j, \rho_{\theta_j})$$

- Apply SLLN to $\sum_{i=1}^n \varphi(X_i, \theta_j, \rho_{\theta_j})$ for each $j = 1, \dots, m$, and use the continuity of $\max$, we have

$$\lim_n \max_{1 \leq j \leq m} \frac{1}{n} \sum_{i=1}^n \varphi(X_i, \theta_j, \rho_{\theta_j}) = \max_{1 \leq j \leq m} E\varphi(X, \theta_j, \rho_{\theta_j}) < \epsilon, \quad \text{a.s.}$$

- Therefore $\lim_n \sup_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n U(X_i, \theta) < \epsilon$, a.s. and thus,

$$\lim_n \sup_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n U(X_i, \theta) \leq 0, \quad \text{a.s.}$$

- Apply the same argument to $-U(x, \theta)$, we can conclude

$$\lim_n \sup_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n -U(X_i, \theta) \leq 0, \quad \text{a.s.}$$

- Note that $0 \leq \sup_{\theta} |g(\theta)| = \max \{ \sup_{\theta} g(\theta), \sup_{\theta} [-g(\theta)] \}$.

Kullback-Leibler Information

- The limit of the log-likelihood ratio is $-E_{\theta_0} \log \frac{f(X|\theta_0)}{f(X|\theta)}$
- Define the **Kullback-Leibler (KL) information number** as

$$K(f_0, f_1) = E_0 \log \frac{f_0(X)}{f_1(X)} = \int \log \left(\frac{f_0(x)}{f_1(x)} \right) f_0(x) d\nu(x)$$

- When f is indexed by θ , we may also write $K(\theta_0, \theta_1)$ for short.

Theorem (Shannon-Kolmogorov Information Inequality)

$K(f_0, f_1) \geq 0$ with equality if and only if $f_1(\omega) = f_0(\omega)$ ν -a.e.

Theorem

Let $X_1, X_2, \dots$ be i.i.d. from P , where P is in the family with density $f(x | \theta)$, $\theta \in \Theta$. Let θ_0 denote the true value of θ .

Suppose

- ① Θ is compact,
- ② $f(x | \theta)$ is continuous in θ for all x ,
- ③ there exists a function $M(x)$ such that $E_{\theta_0}|M(X)| < \infty$ and

$$|\log f(x | \theta) - \log f(x | \theta_0)| \leq M(x), \quad \text{for all } x \text{ and } \theta$$

- ④ (identifiability) $f(x | \theta) = f(x | \theta_0) \nu\text{-a.e.} \Rightarrow \theta = \theta_0$.

Then, for any sequence of maximum-likelihood estimates $\hat{\theta}_n$ of θ ,

$$\hat{\theta}_n \xrightarrow{\text{a.s.}} \theta_0$$

- Let $U(x, \theta) = \log f(x | \theta) - \log f(x | \theta_0)$, and $\mu(\theta) = EU(X, \theta) = -K(\theta_0, \theta) \leq 0$
- Since $f(x | \theta)$ is continuous in θ for all x , so is $U(x, \theta)$.
- Following the proof of the Uniform SLLN, $\mu(\theta)$ is continuous in θ .
- By the identifiability and the Shannon-Kolmogorov Information Inequality, $\mu(\theta) < 0$ if $\theta \neq \theta_0$

- Fixed $\rho > 0$ and $W = \{\theta : \|\theta - \theta_0\| \geq \rho\}$. Then W is compact.
- Apply the Uniform SLLN, we have

$$P \left\{ \lim_{n \rightarrow \infty} \sup_{\theta \in W} \left| \frac{1}{n} \sum_1^n U(X_j, \theta) - \mu(\theta) \right| = 0 \right\} = 1 \quad (2)$$

- Let $\sup_{\theta \in W} \mu(\theta) = -\delta$. Since W is compact and μ is continuous, it achieves its maximal value in W and thus $\delta > 0$
- Denote by Ω_0 the event in Equation (2). When Ω_0 happens, we can choose N large enough such that for any $n \geq N$

$$\sup_{\theta \in W} \frac{1}{n} \sum_1^n U(X_j, \theta) < \sup_{\theta \in W} \mu(\theta) + \delta/2 = -\delta/2$$

- By definition of MLE, $\frac{1}{n} \sum_1^n U(X_j, \hat{\theta}_n) \geq \frac{1}{n} \sum_1^n U(X_j, \theta_0) = 0$. So for $n \geq N$, $\hat{\theta}_n \notin W$, that is, $\|\hat{\theta}_n - \theta_0\| < \rho$.
- Since $P(\Omega_0) = 1$ and ρ is arbitrary, we have $\hat{\theta}_n \xrightarrow{a.s.} \theta_0$

Example: Cauchy Distributions

Suppose X_i 's are i.i.d. from the Cauchy distribution with location parameter θ , whose p.d.f. is

$$f_{\theta}(x) = \pi^{-1} (1 + (x - \theta)^2)^{-1}$$

- Note that $f_{\theta}(x) \leq 1/\pi$ for all θ and x and $\int \frac{\log(1+x^2)}{1+x^2} dx < \infty$, condition (3) holds
- Conditions (2,4) hold.
- If $|\theta_0| \leq A$ is known for some $A > 0$, condition (1) holds with the Euclidean distance.
- Then the MLE restricted on $\Theta_A = [-A, A]$ is strongly consistent by the theorem

Remark 1. On the Continuity

- The assumption that $f(x | \theta)$ is continuous in θ for all x is too restricted.
- Actually, we only need to assume that $f(x | \theta)$ is **upper semicontinuous** (u.s.c.) in θ for all x , which means

$$\lim_{\rho \rightarrow 0} \left\{ \sup_{\|\theta' - \theta\| < \rho} f(x | \theta') \right\} = f(x | \theta)$$

- Accordingly, we need another version of USLLN that only assume $U(x, \theta)$ is upper semicontinuous in θ for any x .
- Instead of getting Eq (1), we can apply the USLLN (u.s.c.) to obtain

$$P_{\theta_0} \left\{ \limsup_{n \rightarrow \infty} \sup_{\theta \in W} \frac{1}{n} \sum_{j=1}^n U(X_j, \theta) \leq \sup_{\theta \in W} \mu(\theta) \right\} = 1$$

The rest of the proof remains the same.

Remark 2. On the Compactness and Metric

When Θ is not compact, we have some remedies

- ① Find a compact set and first show that the estimators eventually lies in this compact set
 - e.g.: If Θ is open and the log likelihood is **concave**, we can focus on $\Theta_r := \overline{B(\theta_0, r)}$ for some $r > 0$.
- ② Replace the space by its suitable compactification
 - In the theorem and proof, we can replace the norm of difference in θ , i.e., $\|\theta - \theta_0\|$ by any metric $d(\theta, \theta_0)$.
 - In this way, the theorems can be extended to a general metric space Θ (in contrary to merely a subset of Euclidean space), and hopefully, Θ will be compact in the new topology induced by the metric
 - Accordingly, the strong consistency is defined as $d(\hat{\theta}_n, \theta_0) \xrightarrow{a.s.} 0$

Consistency of MLEs in General Metric Space

Theorem

Let $X_1, X_2, \dots$ be i.i.d. from P , where P is in the family with density $f(x | \theta)$, $\theta \in \Theta$. Let θ_0 denote the true value of θ .

Suppose

- ① Θ is a compact space with metric $d(\cdot, \cdot)$
- ② $f(x | \theta)$ is upper semicontinuous in θ for all x ,
- ③ there exists a function $M(x)$ such that $E_{\theta_0}|M(X)| < \infty$ and

$$\log f(x | \theta) - \log f(x | \theta_0) \leq M(x), \quad \text{for all } x \text{ and } \theta$$

- ④ for all $\theta \in \Theta$ and sufficiently small $\rho > 0$, $\sup_{d(\theta', \theta) < \rho} f(x | \theta')$ is measurable in x
- ⑤ (identifiability) $f(x | \theta) = f(x | \theta_0) \nu\text{-a.e.} \Rightarrow \theta = \theta_0$.

Then any MLE $\hat{\theta}_n$ of θ is strongly consistent, $d(\hat{\theta}_n, \theta_0) \xrightarrow{a.s.} 0$

Example: Cauchy Distributions (Cont.)

Suppose X_i 's are i.i.d. from the Cauchy distribution with location parameter θ , whose p.d.f. is

$$f_{\theta}(x) = \pi^{-1} (1 + (x - \theta)^2)^{-1}$$

Suppose the parameter space is $\mathbb{R}$. Since it is not compact, we should first compactify it.

- Let's consider an extended space $\bar{\Theta} = \mathbb{R} \cup \{\pm\infty\}$ equipped with metric $d(\theta_1, \theta_2) = |\arg \tan(\theta_1) - \arg \tan(\theta_2)|$ (note that $\arg \tan(\pm\infty) = \pm\pi/2$)
- Further define $\log f(x | -\infty) = -\infty$ and $\log f(x | \infty) = -\infty$ for all x
- Now the conditions in the theorem are satisfied, and we can conclude that $d(\hat{\theta}_n, \theta_0) \xrightarrow{a.s.} 0$
- Since $\theta_0 \neq \pm\infty$ and $\arg \tan$ is 1-1 and continuous, we can conclude that $\hat{\theta}_n \xrightarrow{a.s.} \theta_0$

The theorems here rely on the continuity of p.d.f. but no differentiation is used

- In Theorem 4.17 in the textbook, conditions about the first and second-order derivatives of the log-likelihood are assumed, and the MLE is shown to be weakly consistent

The dominance of the log-likelihood is crucial

- There exists an example where all other conditions are met but the log-likelihood is not dominated by an integrable function, and the MLE is not consistent.

In a particular problem, the consistency of MLEs may also be proved using other methods.

- For instance, when the MLE coincides with the method of moment estimator, we can directly use SLLN and continuous mapping theorem.

M-Estimators

A general method for finding an estimator $\hat{\theta}_n = \hat{\theta}_n(X_1, \dots, X_n)$ is to maximize a criterion function of the type

$$\theta \mapsto S_n(\theta) = \frac{1}{n} \sum_{i=1}^n s_\theta(X_i)$$

Here $s_\theta : \mathcal{X} \mapsto \overline{\mathbb{R}}$ are known functions indexed by θ .

- An estimator maximizing $S_n(\theta)$ over Θ is called an ***M*-estimator**
- Note that an MLE is an *M*-estimator with $s_\theta(x) = \log f(x | \theta)$
- In the theorems of consistency of MLEs, we use $\log f(x | \theta)$ to obtain “ $\mu(\theta) < 0$ for all $\theta \neq \theta_0$ ”.

This can be replaced by “ $E_{\theta_0} s_\theta(X) < E_{\theta_0} s_{\theta_0}(X)$ for all $\theta \neq \theta_0$ ” for general *M*-estimators

Consistency of M -estimators

Theorem

Let S_n be random functions of θ and let S be a fixed function of θ such that

$$\sup_{\theta \in \Theta} |S_n(\theta) - S(\theta)| \xrightarrow{\mathcal{P}} 0,$$

and for every $\rho > 0$,

$$\sup_{\theta: d(\theta, \theta_0) \geq \rho} S(\theta) < S(\theta_0)$$

Then any sequence of estimators $\hat{\theta}_n$ with $S_n(\hat{\theta}_n) \geq S_n(\theta_0) - o_P(1)$ converges in probability to θ_0

- The proof is left for exercise. It is basically the same as the consistency theorem for MLEs
- Condition (6) is the uniform convergence
- Condition (6) is called **well-separation**
- The estimator $\hat{\theta}_n$ in this theorem **nearly maximize** S_n

A Uniform LLN for Lipschitz–Continuous Classes

Let $(X_i)_{i \geq 1}$ be i.i.d. with law P on a measurable space $(\mathcal{X}, \mathcal{A})$. Let $\Theta \subset \mathbb{R}^p$ be nonempty, compact, and let $f : \mathcal{X} \times \Theta \rightarrow \mathbb{R}$ satisfy:

- 1 For each $\theta \in \Theta$, $x \mapsto f(x, \theta)$ is measurable and $Pf(X, \theta)$ exists (finite).
- 2 For each $x \in \mathcal{X}$, the map $\theta \mapsto f(x, \theta)$ is **convex** on Θ .
- 3 (**Lipschitz in θ with integrable modulus**) There exists a measurable $L : \mathcal{X} \rightarrow [0, \infty)$ with $PL(X) < \infty$ such that

$$|f(x, \theta) - f(x, \theta')| \leq L(x) \|\theta - \theta'\| \quad \forall x \in \mathcal{X}, \forall \theta, \theta' \in \Theta.$$

Define the empirical and population functionals

$$P_n f(\cdot, \theta) := \frac{1}{n} \sum_{i=1}^n f(X_i, \theta), \quad Pf(\cdot, \theta) := \int f(x, \theta) dP(x).$$

Then

$$\sup_{\theta \in \Theta} |P_n f(\cdot, \theta) - Pf(\cdot, \theta)| \xrightarrow{\text{a.s.}} 0.$$